No. of Functional Features Included in the Solution

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence Platform (CONSTRUCTAI)	**Maximum Marks:**	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status	Marks Contrib.
1	**Multimodal PPE Detection**	Real-time vision recognition of missing helmets, vests, and harnesses from CCTV/drone images.	Safety Vision Engine (Groq LLaMA 3.2)	**Done**	1.0 Mark
2	**IS 456 Concrete Crack Analyzer**	Structural crack width categorization (hairline vs shear) with Indian Standard repair methods.	Structural QC Diagnostics	**Done**	1.0 Mark
3	**Predictive Weather Risk Advisory**	Doppler rain radar and wind speed threshold alerting to prevent foundation washouts.	Weather Intelligence & API Gateway	**Done**	0.5 Mark
4	**Automated BOQ Material Estimation**	Instant material quantity takeoff generator for cement, aggregates, and rebar.	Project Planning & Takeoff Portal	**Done**	0.5 Mark
5	**Executive PDF Audit Generator**	One-click generation of branded, tamper-proof inspection and compliance audit PDFs.	Reporting & Compliance Engine	**Done**	0.5 Mark
6	**Hazard Sirens & Real-time Alerts**	Multimodal visual warning banners and audible chime alerts for perimeter breach.	Alert & Dispatch Service	**Done**	0.5 Mark
7	**Role-Based Access & Auth (RBAC)**	Secure authentication with role isolation for Inspectors, Engineers, and Project Admins.	Authentication & Session Service	**Done**	0.5 Mark
8	**Offline Fallback Knowledge Cache**	Local indexed storage of IS 456 remediation guidelines when site internet is offline.	Local Storage & Vector Store	**Done**	0.5 Mark

Feature Summary:

Metric	Count / Value
Total Features Planned	**8 Features**
Total Features Implemented	**8 Features (100% Completion)**
Core / Must-Have Features	**6 Features (PPE, Crack, Weather, BOQ, PDF, Auth)**

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Feature Summary (Continued):

Metric	Count / Value
Additional / Nice-to-Have Features	**2 Features (Real-time Sirens & Offline Cache)**
Features Tested & Verified	**8 Features (Verified across all test cases)**

Feature Category Breakdown:

Distribution of functional features across architectural and technical layers of the application.

S.No	Category	Features in Category	Example Features & Implementation Details
1	**User Interface (UI)**	**2 Features**	• Responsive high-contrast outdoor site monitoring dashboard. • Interactive bounding box overlay and structural crack gauge viewer.
2	**Backend / Logic**	**2 Features**	• FastAPI asynchronous processing pipeline. • IS 456:2000 algorithmic crack classification and repair logic.
3	**Database / Storage**	**1 Feature**	• SQLite3 audit logs, site inspection history, and pre-cached local vector JSON standards repository.
4	**API / Integration**	**2 Features**	• Groq Cloud LLaMA 3.2 Vision LPU inference integration. • Open-Meteo Doppler precipitation radar and wind risk sync.
5	**Security / Authentication**	**1 Feature**	• Role-Based Access Control (RBAC) with bcrypt password hashing, JWT token authorization, and zero-data training guarantees.